SWAYAMSHAKTI SAMAL

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PROFESSIONAL SUMMARY

Undergraduate in Mathematical Sciences and Computing with a specialisation in Artificial Intelligence. My academic focus centers on mathematics, statistics, and computing, particularly how they apply to financial markets. Through my university coursework and training in financial engineering and algorithmic trading, I have built a strong interest in quantitative research and systematic approaches to studying market behaviour. I am particularly interested in problems that bring together mathematical reasoning, programming, and real-world applications.

EDUCATION

Sri Sathya Sai Institute of Higher Learning (SSSIHL) 2023 - 2027
B.S. (Hons.) in Mathematical Sciences and Computing
Specialization in Artificial Intelligence | **Grade: A+**

Relevant Coursework: Probability and Distributions, Statistical Inference, Advanced Linear Algebra, Real Analysis, Numerical Analysis, Optimization Techniques for Machine Learning, Differential Equations, Mathematical Modelling, Data Structures and Algorithms, Database Management Systems, and Data Mining & Machine Learning.

PROFESSIONAL EXPERIENCE

Software Development Intern | Sri Sathya Sai Vidya Vahini 2024 - 2025
Puttaparthi, India

- Developed an interactive browser-based educational game using JavaScript, HTML, and CSS, implementing core game logic including drag-and-drop mechanics, collision detection, and level progression.
- Built a real-time mouse tracking system to capture user behavior telemetry (clicks, cursor movement, timestamps), processing and exporting interaction logs for usability and performance analysis using DOM manipulation and event handling.

PROJECTS

SPY Variance Risk Premium: SVI Calibration, GJR-GARCH & Delta-Hedged Backtesting

- **Large-Scale Options Data Engineering:** Built a chunked ingestion and static no-arbitrage filtering pipeline in Python to process **5.39 million historical SPY option quotes (1.25 GB, 2019–2023)**, scrubbing crossed order books and aligning inline spot returns across **10.15 million** clean Call/Put records.
- **Volatility Surface & Asymmetric Forecasting:** Extracted risk-neutral implied variance (Q-measure) via Black-Scholes Newton-Raphson inversion and bounded **SVI smile calibration**, comparing it against out-of-sample physical variance (P-measure) forecasts from a 252-day rolling **GJR-GARCH(1,1)** model with Student-t innovations.
- **Delta-Hedged Backtesting & Friction Analysis:** Simulated **180 short ATM straddles** across **1,001 out-of-sample trading days** with daily closing delta rebalancing, demonstrating positive gross alpha (+$2,095, -0.40% max drawdown) while quantifying how bid-ask spread crossing and slippage compress net returns to break-even (-0.01%).

FINANCE TRAINING AND CERTIFICATIONS

1. Financial Engineering and Risk Management Specialization – Columbia University
   https://www.coursera.org/account/accomplishments/specialization/24Z2NPMM6D9L
2. Python for Finance & Algorithmic Trading — Udemy
   https://www.udemy.com/certificate/UC-5965465f-37cf-4a63-96e5-1be6a0a8d74e/

SKILLS

Mathematics: Probability&Statistics, LinearAlgebra,Calculus,Optimization, NumericalAnalysis.

Programming & Data: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Plotly, SQL, C/C++, Microsoft Excel

Quantitative Finance: Financial Engineering, Investment Management, Model Optimization, Asset Management, Regression Analysis, Derivatives, Risk Management, Backtesting,

Professional: Analytical Thinking, Problem-Solving, Teamwork, Attention to Detail.